

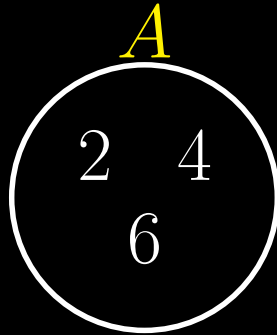
Representing Sets and Operations Using Venn Diagrams

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A **Venn diagram** is a picture that shows the relationship between sets.

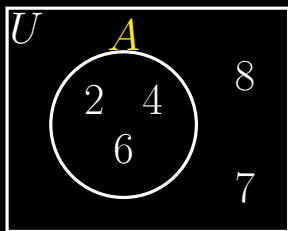
Sets and Elements

A **set** is represented by a closed curve, while its **elements** are written inside the curve.



Universal Set

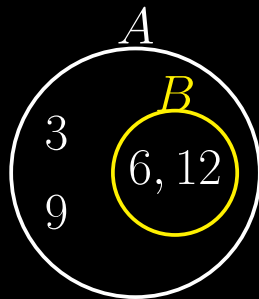
The **universal set**, denoted by U , contains all elements under consideration. It is drawn as a rectangle.



Elements of U that are not in A are written outside the circle.

A **subset** is a set whose elements are all contained in another set.

Subset Diagram

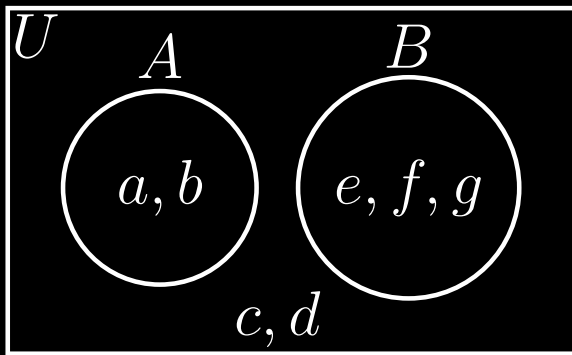


$$B \subseteq A$$

The circle for B is inside the circle for A .

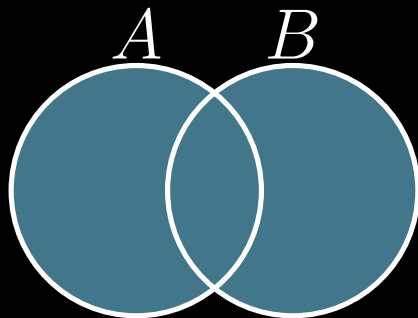
Disjoint Sets

Two sets with no common elements are **disjoint**. Their circles do not overlap.



The **union** of two sets, written as $A \cup B$, consists of all elements that belong to A , to B , or to both.

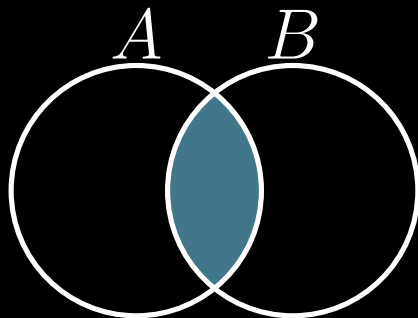
Union Diagram



$$A \cup B$$

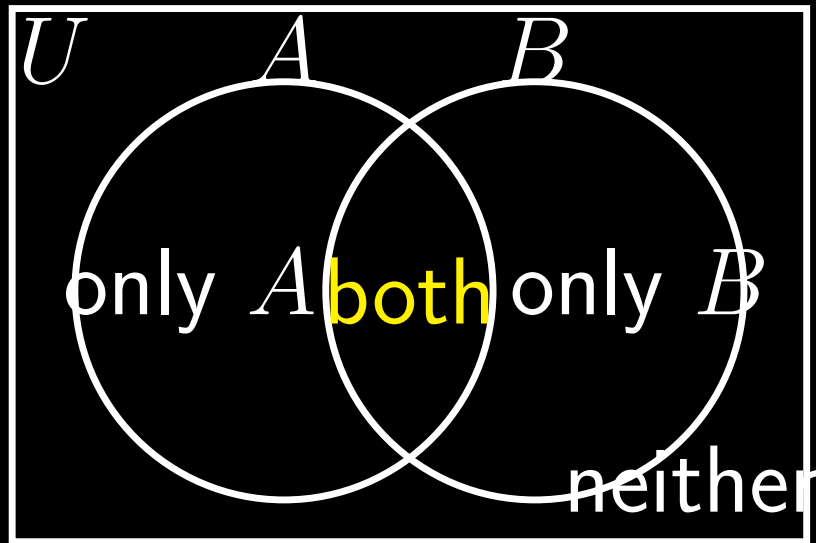
The **intersection** of two sets, written as $A \cap B$, consists of all elements common to both sets.

Intersection Diagram



$$A \cap B$$

Regions of a Venn Diagram



Steps in Drawing a Venn Diagram

1. Draw a rectangle for U .
2. Draw a circle for each set.
3. Write the elements in **both** sets in the overlap.
4. Write the elements in **only one** set in its own circle.
5. Write the remaining elements of U outside the circles.

Memory Tip

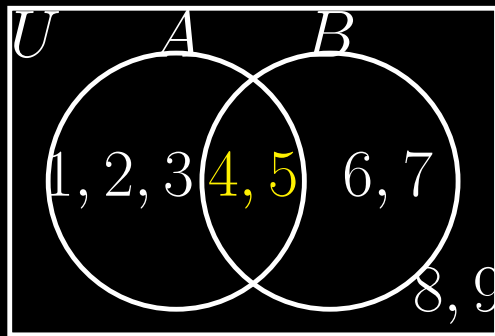
Middle First!

Fill the overlap first. Then fill the rest of each circle.

Examples

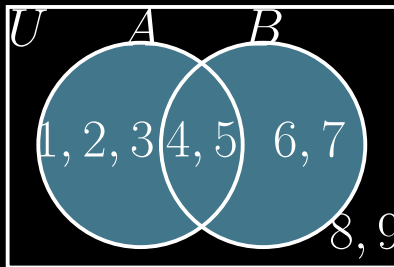
Example 1

Let $U = \{1, 2, \dots, 9\}$, $A = \{1, 2, 3, 4, 5\}$,
 $B = \{4, 5, 6, 7\}$. Draw a Venn diagram.



Example 2

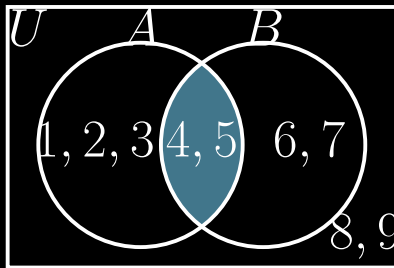
Using the sets in Example 1, shade $A \cup B$.



$$A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$$

Example 3

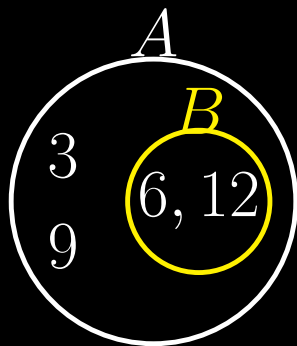
Using the sets in Example 1, shade $A \cap B$.



$$A \cap B = \{4, 5\}$$

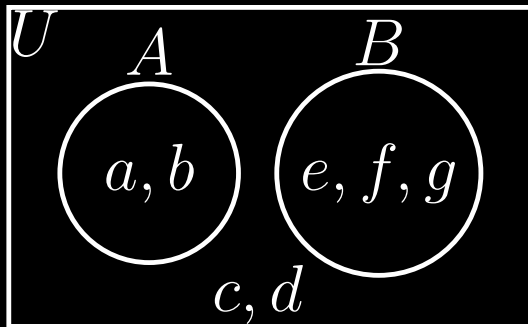
Example 4

Let $A = \{3, 6, 9, 12\}$ and $B = \{6, 12\}$. Draw a Venn diagram to show that $B \subseteq A$.



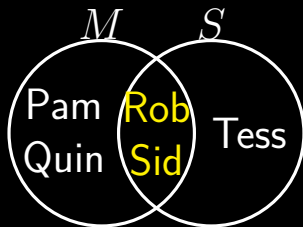
Example 5

Let $U = \{a, b, c, d, e, f, g\}$, $A = \{a, b\}$, $B = \{e, f, g\}$. Illustrate the two disjoint sets.



Example 6

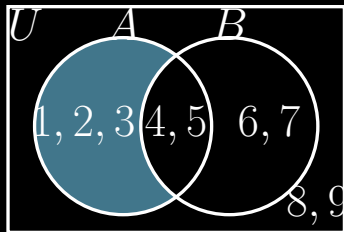
In a class, $M = \{\text{Pam, Quin, Rob, Sid}\}$, $S = \{\text{Rob, Sid, Tess}\}$. Draw a Venn diagram and determine $M \cap S$.



$$M \cap S = \{\text{Rob, Sid}\}$$

Example 7

Using the sets in Example 1, identify the elements belonging **only** to A .



$\{1, 2, 3\}$

Practice Exercises

Practice 1

Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $A = \{2, 4, 6, 8, 10\}$, $B = \{1, 2, 3, 4, 5\}$. Draw a Venn diagram and shade $A \cup B$.

Practice 2

Using the sets in Exercise 1, draw a Venn diagram and shade $A \cap B$.

Practice 3

Let $U = \{a, b, c, d, e, f, g, h\}$, $A = \{a, c, e, g\}$, $B = \{c, d, e, f\}$. Illustrate the sets using a Venn diagram and list the elements in each region.

Practice 4

Let $A = \{2, 4, 6, 8\}$, $B = \{2, 4\}$. Draw a Venn diagram to show that $B \subseteq A$.

Practice 5

Let $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$, $A = \{1, 2, 3, 4\}$, $B = \{5, 6, 7\}$. Illustrate the two disjoint sets using a Venn diagram.

Practice 6

In a class, $M = \{\text{Ana, Ben, Carl, Dana}\}$, $S = \{\text{Carl, Dana, Ella}\}$. Draw a Venn diagram and determine $M \cap S$.

Practice 7

In a survey, $F = \{\text{apple, banana, mango}\}$, $G = \{\text{banana, mango, orange}\}$. Draw a Venn diagram and determine $F \cup G$.

Practice 8

Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, $A = \{2, 4, 6, 8\}$, $B = \{4, 5, 6, 7\}$. Illustrate the Venn diagram and identify the elements belonging only to A .

Practice 9

Let $U = \{a, b, c, d, e, f, g\}$, $A = \{a, b, c, d\}$, $B = \{c, d, e, f\}$. Illustrate the Venn diagram and identify the elements belonging only to B .

Practice 10

Let $U = \{1, 2, \dots, 12\}$, $A = \{2, 4, 6, 8, 10, 12\}$, $B = \{3, 6, 9, 12\}$. Draw a Venn diagram and identify:

(a) $A \cap B$

(b) $A \cup B$

Activities

Activity 1

Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $A = \{1, 3, 5, 7, 9\}$, $B = \{5, 6, 7, 8, 9\}$. Draw a Venn diagram and shade $A \cup B$.

Activity 2

Using the sets in Activity 1, draw a Venn diagram and shade $A \cap B$.

Activity 3

Let $U = \{m, n, o, p, q, r, s, t\}$, $A = \{m, o, q, s\}$, $B = \{o, p, q, r\}$. Illustrate the sets using a Venn diagram and list the elements in each region.

Activity 4

Let $A = \{5, 10, 15, 20\}$, $B = \{10, 20\}$. Draw a Venn diagram to show that $B \subseteq A$.

Activity 5

Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, $A = \{1, 2, 3\}$, $B = \{6, 7, 8\}$. Illustrate the two disjoint sets using a Venn diagram.

Activity 6

In a class, $M = \{\text{Leo, Mia, Noah, Pia}\}$, $S = \{\text{Noah, Pia, Ryan}\}$. Draw a Venn diagram and determine $M \cap S$.

Activity 7

In a survey, $F = \{\text{grapes, apple, orange}\}$, $G = \{\text{orange, melon, apple}\}$. Draw a Venn diagram and determine $F \cup G$.

Activity 8

Let $U = \{10, 11, 12, 13, 14, 15, 16, 17, 18\}$, $A = \{10, 12, 14, 16\}$, $B = \{12, 13, 14, 15\}$. Illustrate the Venn diagram and identify the elements belonging only to A .

Activity 9

Let $U = \{h, i, j, k, l, m, n\}$, $A = \{h, i, j, k\}$, $B = \{j, k, l, m\}$. Illustrate the Venn diagram and identify the elements belonging only to B .

Activity 10

Let $U = \{1, 2, \dots, 15\}$, $A = \{5, 10, 15\}$, $B = \{3, 6, 9, 12, 15\}$. Draw a Venn diagram and identify:

(a) $A \cap B$

(b) $A \cup B$

Thank you!